Target Reference

What are the prediction targets? / Quelles sont les cibles ?

Project / Projet: Harvest window prediction - soft winter wheat (BTH)
Region / Region: Centre-Val de Loire, France (28, 45, 41, 37, 18, 36)
Period / Periode: 2020-2024 | **Sample:** 1,500 parcels (300/yr)
Author / Auteur: Regis Kassi |
Scope: defines the targets used in the modeling phase / definit les cibles utilisees en phase de modelisation

This document is bilingual: each section gives the English version first (tagged EN), then the French version (tagged FR). / Ce document est bilingue : chaque section donne d'abord l'anglais (EN), puis le francais (FR).

1. Two targets at a glance / Les deux cibles en un coup d'oeil

EN

The project defines **two** harvest-date targets. Both express the harvest day-of-year (DOY) of soft winter wheat, but at different granularities.

FR

Le projet definit **deux** cibles de date de recolte. Les deux expriment le jour de l'annee (DOY) de la recolte du ble tendre, mais a des granularites differentes.

Aspect	Regional target	Derived parcel-level target
Column / Colonne	harvest_doy	harvest_doy_derived
Granularity	1 value per year	1 value per parcel
Distinct values	5 (2020-2024)	142
Source	CereObs / FranceAgriMer	S2 senescence + S1 recovery, calibrated
Reliability	Official, high	Calibrated proxy (not field truth)
Inter-parcel var.	None	Yes (intra-year std ~9 days)
Use / Usage	Baseline	Main / operational

2. Regional target / Cible regionale

EN

Extracted from the CereObs/FranceAgriMer weekly harvest-progress curve by interpolating the date at which 50% of the regional area is harvested. One value per year, shared by all parcels of that year. Reliable but flat (no inter-parcel signal).

FR

Extraite de la courbe hebdomadaire de progression de la recolte CereObs/FranceAgriMer, en interpolant la date a laquelle 50% de la surface regionale est recoltee. Une valeur par an, partagee par toutes les parcelles de l'annee. Fiable mais plate (aucun signal entre parcelles).

Regional anchors / Ancres regionales (validated, 0-day error):

Year	harvest_doy	Date (approx.)	Climatic note / Note climatique
2020	185	~3 Jul	Normal
2021	199	~18 Jul	Cool spring / printemps frais (+14 d)
2022	183	~2 Jul	May heatwave / canicule mai (early)
2023	184	~3 Jul	Normal
2024	195	~13 Jul	Wet spring / printemps humide

3. Derived parcel-level target / Cible parcellaire derivee

EN

One estimated harvest DOY per parcel, built per parcel-year by: (1) extracting the Sentinel-2 senescence DOY (NDVI drop) and the Sentinel-1 recovery DOY (VV/VH ratio rising after the canopy minimum); (2) quality-weighted fusion of the two signals; (3) annual offset calibration so each year's parcel median equals the regional anchor; (4) biological clipping to [165,225] DOY; (5) regional fallback when both signals are missing.

FR

Une date de recolte estimee par parcelle, construite par parcelle-annee via : (1) extraction du DOY de senescence Sentinel-2 (chute du NDVI) et du DOY de remontee Sentinel-1 (ratio VV/VH qui remonte apres le minimum de canopee) ; (2) fusion ponderee par la qualite des deux signaux ; (3) calibration annuelle par decalage pour que la mediane des parcelles egale l'ancre regionale ; (4) clipping biologique a [165,225] DOY ; (5) repli sur la valeur regionale si les deux signaux manquent.

Traceability columns / Colonnes de tracabilite:

Column	Type	Values	Description
harvest_doy_derived	int	[165,225]	Derived harvest DOY / DOY derive
signal_source	string	both / s2_only / s1_only / fallback	Which signals used
was_clipped	bool	true / false	Value clipped to bounds

Signal sources: both=70.3%, s2_only=26.7%, s1_only=1.6%, fallback=1.4%. 47 parcels (3.1%) clipped. Intra-year std ~9 days; inter-annual corr with regional means = 1.000.

4. Important: proxy nature / Important : nature de proxy

EN - The derived target is a **calibrated multi-sensor proxy**, NOT field ground truth. No in-situ harvest dates were measured. Any modeling claim must state this; performance is measured against an estimated signal, not observed harvest.

FR - La cible derivee est un **proxy multi-capteurs calibre**, et NON une verite terrain. Aucune date de recolte n'a ete mesuree sur le terrain. Toute affirmation de modelisation doit le preciser ; la performance est mesuree contre un signal estime, pas une recolte observee.

5. ML formulations / Formulations ML

EN

The same DOY target can be modeled three ways:

- **Regression** - predict harvest_doy as a continuous value. Primary metric: MAE in days.
- **Nominal classification** - bin the DOY into date ranges (classes) and predict the class. References: Frank & Hall 2001; Vannoppen 2022.
- **Ordinal regression** - treat the date ranges as ordered (a 'sweet-spot' window), respecting that adjacent classes are closer than distant ones.

FR

La meme cible DOY peut etre modelisee de trois facons :

- **Regression** - predire harvest_doy en valeur continue. Metrique : MAE en jours.
- **Classification nominale** - regrouper le DOY en plages de dates (classes) et predire la classe. References : Frank & Hall 2001 ; Vannoppen 2022.
- **Regression ordinale** - traiter les plages comme ordonnees (fenetre 'sweet-spot'), en respectant que les classes voisines sont plus proches que les eloignees.

6. Which target in which dataset / Quelle cible dans quel jeu

Dataset	Target column	Notes
master_raw_regional	harvest_doy	Regional, raw exploration
master_ml_regional	harvest_doy	Regional baseline
master_raw_derived	harvest_doy_derived	Derived, raw exploration
master_ml_derived_may31	harvest_doy_derived	Operational, ~6 wk lead
master_ml_derived_june15	harvest_doy_derived	Operational, ~4 wk lead

EN

Recommendation: use the derived cutoff datasets for the main results and the regional ML master as a baseline. Validate with Leave-One-Year-Out (5 folds).

FR

Recommandation : utiliser les jeux derives a cutoff pour les resultats principaux et le master ML regional comme baseline. Valider en Leave-One-Year-Out (5 plis).